

Oxytocin and analgesia: future trends

Abimael González-Hernández, Gerardo Rojas-Piloni, and Miguel Condés-Lara

Departamento de Neurobiología del Desarrollo y Neurofisiología, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Campus UNAM-Juriquilla, Boulevard Juriquilla 3001, Queretaro, Qro. 76230, Mexico

Although best known for its function in lactation, uterine contraction during parturition, and for its role in social behaviour, the neurohormone oxytocin (OT) has also been implicated as an important mediator in endogenous analgesia. Electrophysiological experiments show that OT is able to block the activity of nociceptive A δ - and C-fibres without affecting non-nociceptive fibres (A β -fibres) [1,2]. Indeed, OT could exert its antinociceptive effects at spinal [3–5], supraspinal [6], and at peripheral levels [3,7,8] (Figure 1). Despite the potential therapeutic use of this neurohormone [9], the identity of the main receptor mediating inhibition of nociception remains obscure; some studies [2–4,7,11] suggest that oxytocin receptors (OTRs) are responsible, but others [8,12] favoured the role of vasopressin 1a receptors (V1aRs). In this context, a key question is whether OTRs or V1aRs may be involved in OT-induced antinociception.

Currently, we know that OT is able to reduce pain transmission by activation of several pathways. At spinal levels, OT enhances α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (GABA) release from spinal interneurons [1] located in the substantia gelatinosa [4]. Furthermore, OT reduces the GABA-evoked Ca²⁺ transients in trigeminal nociceptive neurons; an effect blocked by atosiban (OTR-preferring antagonist) in rats [10]. In addition, in small- and medium-size dorsal root ganglion (DRG) neurons, the OT-induced inhibition of intracellular Ca²⁺ currents was blocked by desGly-NH₂-d(CH₂)₅[D-Tyr²,Thr⁴]OVT (an OTR selective antagonist), but not by d(CH₂)₅[Tyr(me)²,Arg⁸]VP (a V1aR selective antagonist), suggesting a role for OTRs [2]. Interestingly, a recent study in these type of cells showed that SR 49059 (a potent and selective V1aR receptor antagonist) was able to block OT-induced inhibition of acid-sensing ion channels (ASICs), supporting a role for V1aRs [8]. Together, these results indicate that, depending on the compounds used, the receptor involved in OT-induced analgesia could be the OTR or the V1aR.

At the spinal level, the role of OTR is strongly supported, considering that spinal administration of OT, but not vasopressin (AVP) (at doses ranging from 10⁻⁵ to 10⁻¹⁰ M), is able to selectively decrease the nociceptive neuronal activity of A δ - and C-fibres [3]. In addition, since spinal administration of d(CH₂)₅[Tyr(Me)²,Thr⁴,Tyr-NH₂⁹]OVT (OTR-preferring antagonist) reversed the antinociceptive effect of OT or electrical stimulation of the

paraventricular hypothalamic nuclei (PVN), the role of spinal OTR is reinforced [1,3]. Accordingly, in whole-cell voltage-clamp recordings in neurons of the substantia gelatinosa, where OT enhanced the GABAergic inhibitory transmission, d(CH₂)₅¹-[Tyr(Me)² Thr⁴, Orn⁸, des-Gly-NH₂⁹]-vasotocin (dVOT; OTR-preferring antagonist) was able to block the OT effect [4].

By contrast, in analgesia at the periphery, some studies show involvement of OTRs [2,7,11], whereas others favoured the V1aR [8,12]. Using extracellular unitary recordings of wide dynamic range neurons, Juif and Poisbeau [11] showed in anaesthetized rats that a physiological dose of intravenous (i.v.) [Thr⁴, Gly⁷]-oxytocin 5 ng (TGOT; OTR agonist) reduces C fibre activity, whereas a supra-maximal dose of TGOT (5 μ g; i.v.) or 50 ng AVP increases the C-fibre discharge, suggesting an unspecific effect in this instance. Accordingly, the antinociceptive effect of TGOT was blocked by i.v. dVOT (50 μ g) and the pronociceptive effect of TGOT and AVP was abolished by the [phenylacetyl¹, O-Me-D-Tyr², Arg^{6,8}, Lys⁹]-vasopressin amide (a V1aR-preferring antagonist; 50 μ g). These results support not only the role of OTR in OT-induced analgesia, but also the pronociceptive role of V1aR peripheral receptors. At this point, it is important to keep in mind that since the low dose of OT is unable to cross the blood–brain barrier, the effect of this peptide must be related to a peripheral mechanism. Similarly, in a model of chronic abdominal pain in mice where OTR expression is enhanced in colonic DRG, it has been shown (*in vitro* and *in vivo*) that seleno-oxytocin analogues inhibit the neuronal activity of colonic nociceptors [7], suggesting the role of OTR.

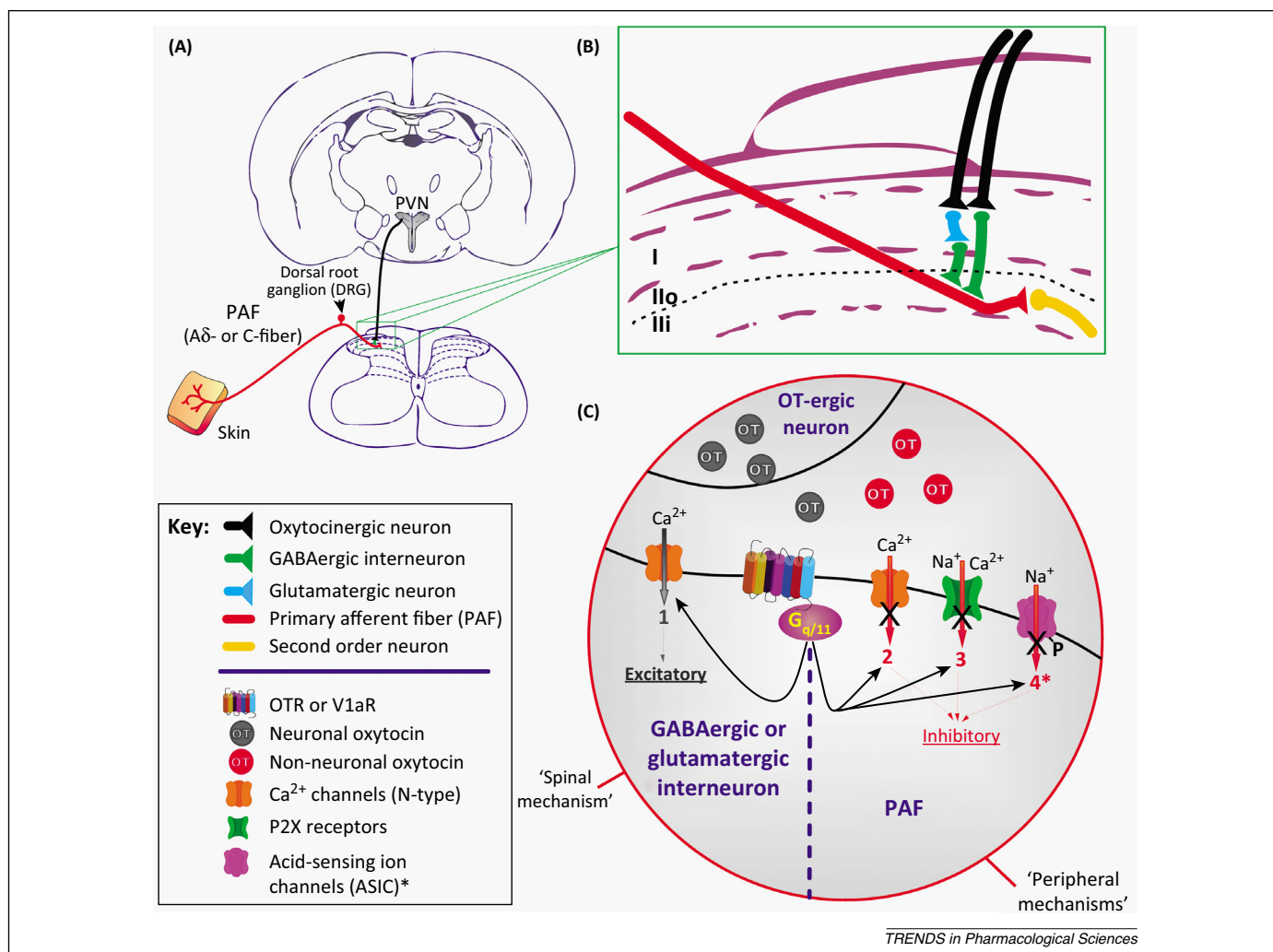
By contrast, using whole-cell patch clamp and voltage clamp, Qiu *et al.* [8] showed that OT inhibited the ASIC currents activated by acidification of the tissue in isolated small and medium diameters of DRG neurons in a dose-dependent fashion. This effect was mimicked by AVP at the same effective dose (10⁻⁵ M). Interestingly, since this effect was blocked by SR 49059 but not by L-368,899 (a potent and selective OTR receptor antagonist), and coupled to the fact that, in mice lacking V1aR, OT and AVP lost their antinociceptive capacity (as previously reported [12]), the authors claimed that OT-induced antinociception is mediated by V1aR. This statement is reinforced by behavioural experiments in rats: the flinching behaviour induced by acetic acid was inhibited by intraplantar OT or AVP, an effect blocked by SR 49059. Taken together, the above data strongly support the role of peripheral V1aR in OT-induced antinociception. However, it is important to keep in mind that although V1aR knockout mice support the role of V1aR, the potential limitations (e.g., the presence of compensation mechanisms to adjust the absence of the

Corresponding author: González-Hernández, A. (abimaelgh@gmail.com).

Keywords: oxytocin; vasopressin; pain; analgesia.

0165-6147/

© 2014 Elsevier Ltd. All rights reserved. <http://dx.doi.org/10.1016/j.tips.2014.09.004>



deleted receptors) using transgenic mice could not be avoided. Furthermore, the effect of L-368,899 was not tested in the behavioural experiments. In both cases, the lack of a well-defined pharmacological approach to validate the receptor involved remains a concern that needs to be clarified in the future.

OT emerges as a novel analgesic to treat pain conditions in humans [13], but the receptor involved remains elusive. Currently, the peripheral contribution of V1aR or OTR cannot be categorically ruled out, and we cannot exclude a possible differential participation of these receptors at different levels (i.e., peripheral versus spinal and supraspinal levels). Indeed, research about the receptor(s) involved in the antinociceptive effects of OT at the peripheral ending of nociceptors and supraspinal levels remains the central question to be answered.

Acknowledgements

This work was supported by National Council of Science and Technology (CONACyT – Mexico, Grant No. 164536).

References

- Rojas-Piloni, G. *et al.* (2007) GABA-mediated oxytocinergic inhibition in dorsal horn neurons by hypothalamic paraventricular nucleus stimulation. *Brain Res.* 1137, 69–77
- Hobo, S. *et al.* (2012) Oxytocin inhibits the membrane depolarization-induced increase in intracellular calcium in capsaicin sensitive sensory neurons: a peripheral mechanism of analgesic action. *Anesth. Analg.* 114, 442–449
- Rojas-Piloni, G. *et al.* (2010) Oxytocin, but not vasopressin, modulates nociceptive responses in dorsal horn neurons. *Neurosci. Lett.* 476, 32–35
- Jiang, C-Y. *et al.* (2014) Synaptic modulation and inward current produced by oxytocin in substantia gelatinosa neurons of adult rat spinal cord slices. *J. Neurophysiol.* 111, 991–1007

- 5 Breton, J.D. *et al.* (2008) Oxytocin-induced antinociception in the spinal cord is mediated by a subpopulation of glutamatergic neurons in laminae I–II which amplify GABAergic inhibition. *Mol. Pain* 4, 19
- 6 Russo, R. *et al.* (2012) Central administration of oxytocin reduces hyperalgesia in mice: implications for cannabinoid and opioid systems. *Peptides* 38, 81–88
- 7 de Araujo, A.D. *et al.* (2014) Selenoether oxytocin analogues have analgesic properties in a mouse model of chronic abdominal pain. *Nat. Commun.* 5, 3165
- 8 Qiu, F. *et al.* (2014) Oxytocin inhibits the activity of acid-sensing ion channels through the vasopressin, V1A receptor in primary sensory neurons. *Br. J. Pharmacol.* 171, 3065–3076
- 9 Rash, J.A. *et al.* (2014) Oxytocin and pain: a systematic review and synthesis of findings. *Clin. J. Pain* 30, 453–462
- 10 Mazzuca, M. *et al.* (2011) Newborn analgesia mediated by oxytocin during delivery. *Front. Cell. Neurosci.* 5, 3
- 11 Juif, P.E. and Poisbeau, P. (2013) Neurohormonal effects of oxytocin and vasopressin receptor agonists on spinal processing in male rats. *Pain* 154, 1449–1456
- 12 Schorscher-Petcu, A. *et al.* (2010) Oxytocin-induced analgesia and scratching are mediated by the vasopressin-1A receptor in the mouse. *J. Neurosci.* 30, 8274–8284
- 13 Yaksh, T.L. *et al.* (2014) Preclinical toxicity screening of intrathecal oxytocin in rats and dogs. *Anesthesiology* 120, 951–961